

FIT5032 eFolio Evidence Report

Tasks 3 and 4

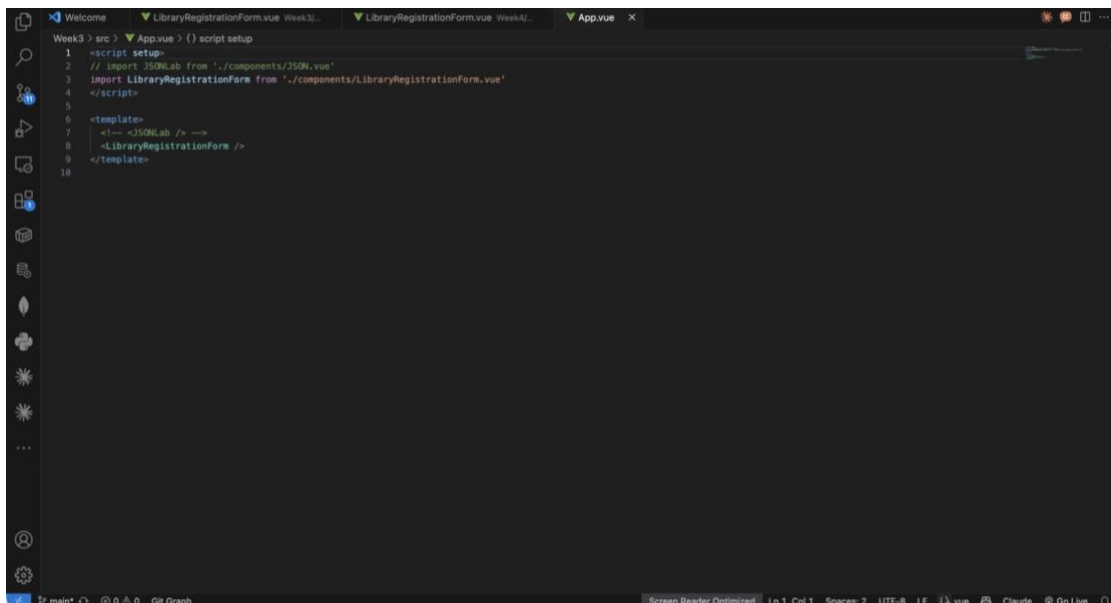
Student: He Li

Student ID: 35381043

Weeks: Week 3 and Week 4

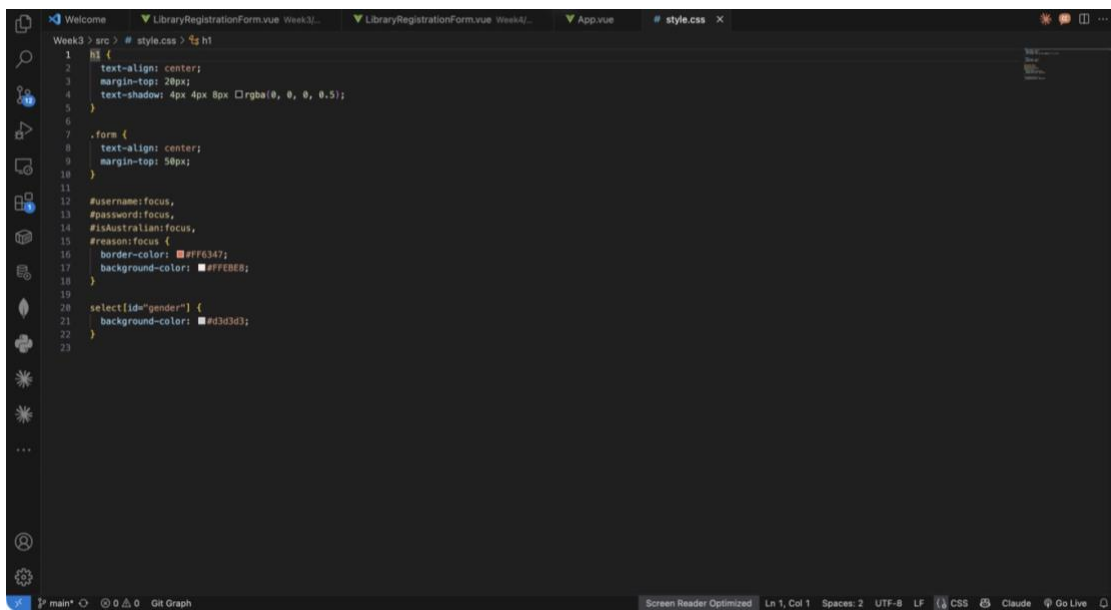
1. Week 3 - Component, HTML and CSS

I created the LibraryRegistrationForm component and displayed it from App.vue. I added the form fields and used the CSS selectors from the studio activity.



```
Week3 > src > App.vue > () script setup
1 <script setup>
2 // import JSONLab from './components/JSONLab.vue'
3 import LibraryRegistrationForm from './components/LibraryRegistrationForm.vue'
4 </script>
5
6 <template>
7   <!-- <JSONLab /> -->
8   <LibraryRegistrationForm />
9 </template>
10
```

Figure 1. Activity 1 component import and display in App.vue.

A screenshot of the Visual Studio Code editor interface. The top bar shows the file explorer with 'style.css' selected. The editor window displays CSS code for a registration form. The code includes a 'h1' selector for the main heading, a '.form' selector for the form container, and several input selectors for 'username', 'password', 'isAustralian', and 'reason'. It also includes a 'select' selector for the 'gender' dropdown. The code is as follows:

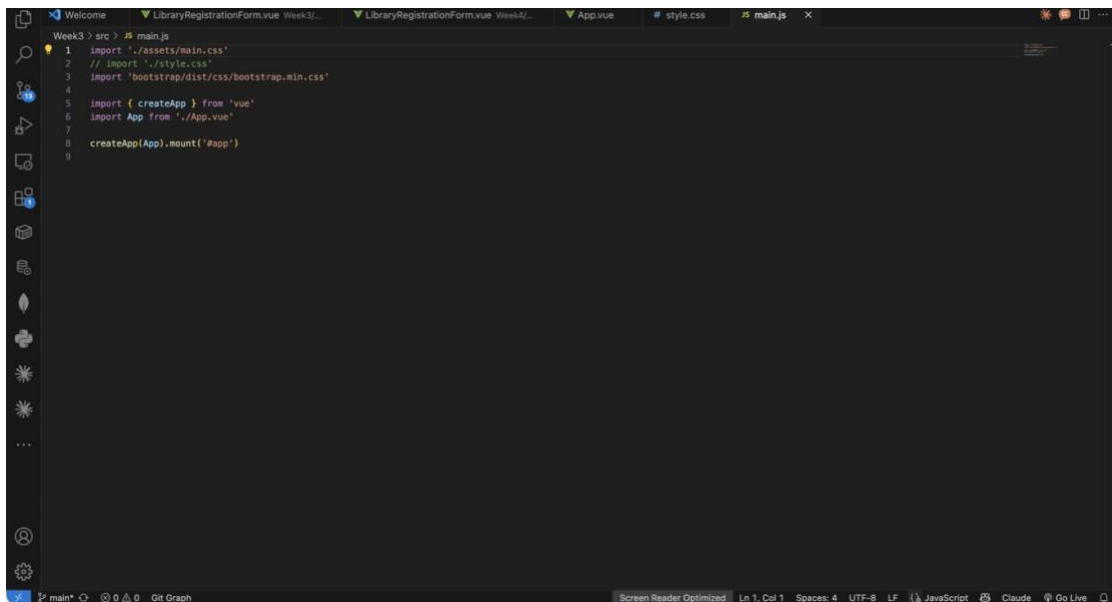
```
1 h1 {
2   text-align: center;
3   margin-top: 20px;
4   text-shadow: 4px 4px 8px rgba(0, 0, 0, 0.5);
5 }
6
7 .form {
8   text-align: center;
9   margin-top: 50px;
10 }
11
12 #username:focus,
13 #password:focus,
14 #isAustralian:focus,
15 #reason:focus {
16   border-color: #FF6347;
17   background-color: #FFEEB8;
18 }
19
20 select[id="gender"] {
21   background-color: #d3d3d3;
22 }
23
```

The status bar at the bottom indicates 'Screen Reader Optimized', 'Ln 1, Col 1', 'Spaces: 2', 'UTF-8', 'LF', and 'CSS'.

Figure 2. Activity 3 tag, class, ID and attribute selectors.

2. Week 3 - Bootstrap Form

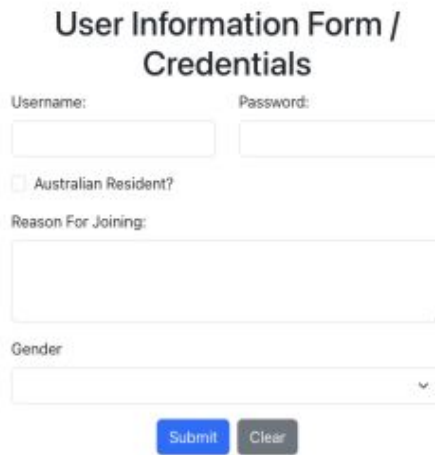
I imported Bootstrap CSS in main.js and used Bootstrap classes to arrange the form.

A screenshot of the Visual Studio Code editor interface. The top bar shows the file explorer with 'main.js' selected. The editor window displays JavaScript code for importing Bootstrap CSS and creating a Vue application. The code is as follows:

```
1 import './assets/main.css'
2 // import './style.css'
3 import 'bootstrap/dist/css/bootstrap.min.css'
4
5 import { createApp } from 'vue'
6 import App from './App.vue'
7
8 createApp(App).mount('#app')
9
```

The status bar at the bottom indicates 'Screen Reader Optimized', 'Ln 1, Col 1', 'Spaces: 4', 'UTF-8', 'LF', and 'JavaScript'.

Figure 3. Activity 4 Bootstrap CSS import.



The image shows a web form titled "User Information Form / Credentials". It contains the following fields and controls:

- Username:** A text input field.
- Password:** A text input field.
- Australian Resident?:** A checkbox.
- Reason For Joining:** A large text area.
- Gender:** A dropdown menu.
- Submit:** A blue button.
- Clear:** A grey button.

Figure 4. Activity 5 Bootstrap form on a wider screen.

3. Submitted User Card (Week 3)

I used v-model to store the form values. After I click Submit, the entered information is shown in a Bootstrap card.

User Information Form / Credentials

Username:	heli_student	Password:	
☒ Australian Resident?			
Reason For Joining:			
I want to use the library resources for my study.			
Gender			
Other			
Submit		Clear	

User Information
Username: heli_student
Password: Example1!
Australian Resident: Yes
Gender: other
Reason: I want to use the library resources for my study.



Figure 5. Activity 6 submitted user information card.

4. Responsive Breakpoint (Week 3)

I changed the input columns to col-sm-6. The username and password fields stay in two columns at 600px, while a very small screen can stack them.



User Information Form / Credentials

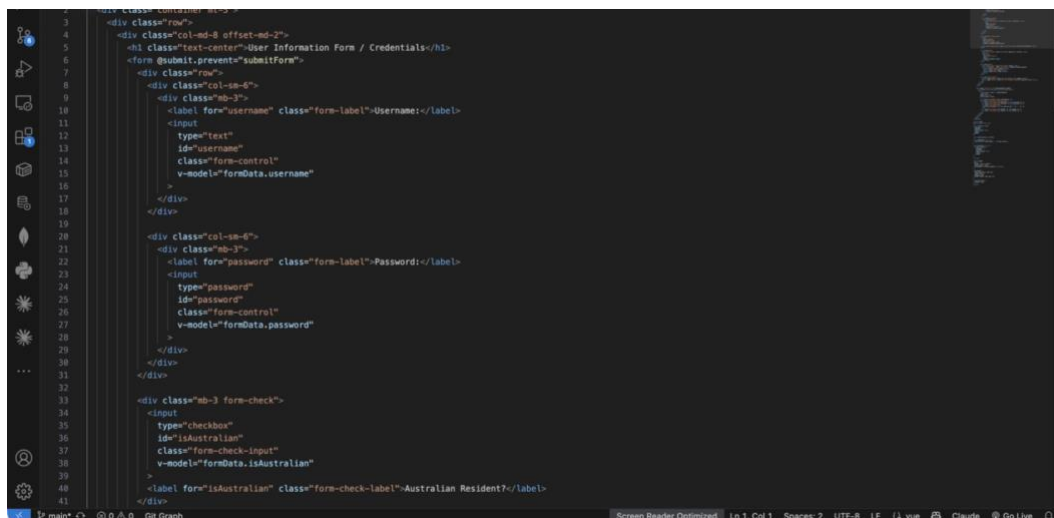
Username: Password:

☐ Australian Resident?

Reason For Joining:

Gender

Figure 6. Activity 7 form tested at a 600px viewport.



```
1 <div class="container">
2   <div class="row">
3     <div class="col-md-8 offset-md-2">
4       <h1 class="text-center">User Information Form / Credentials</h1>
5       <form @submit.prevent="submitForm">
6         <div class="row">
7           <div class="col-sm-6">
8             <div class="mb-3">
9               <label form="username" class="form-label">Username</label>
10              <input
11                type="text"
12                id="username"
13                class="form-control"
14                v-model="formData.username"
15              >
16            </div>
17          </div>
18          <div class="col-sm-6">
19            <div class="mb-3">
20              <label form="password" class="form-label">Password</label>
21              <input
22                type="password"
23                id="password"
24                class="form-control"
25                v-model="formData.password"
26              >
27            </div>
28          </div>
29        </div>
30        <div class="mb-3 form-check">
31          <input
32            type="checkbox"
33            id="isAustralian"
34            class="form-check-input"
35            v-model="formData.isAustralian"
36          >
37          <label form="isAustralian" class="form-check-label">Australian Resident</label>
38        </div>
39        <div class="mb-3">
40          <div class="form-label">Reason For Joining</div>
41          <input type="text">
42        </div>
43        <div class="mb-3">
44          <div class="form-label">Gender</div>
45          <input type="text">
46        </div>
47      </form>
48    </div>
49  </div>
50 </div>
```

Figure 7. Bootstrap sm breakpoint used for the two input columns.

5. Week 4 - Form Validation Result

I added validation for username, password, Australian residency, reason for joining and gender. Invalid fields show Bootstrap error styles and a clear message.

The screenshot shows a web form titled "User Information Form / Credentials". It contains the following fields and validation messages:

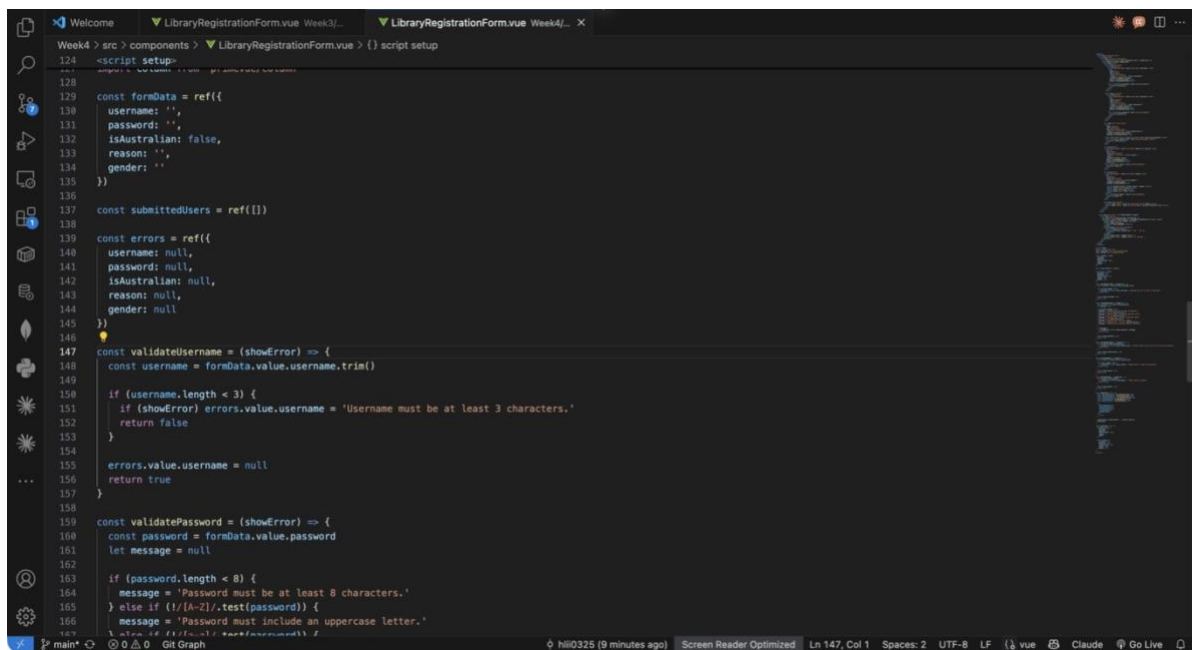
- Username:** A text input field with a red border and a red error icon. Below it, the message "Username must be at least 3 characters." is displayed in red.
- Password:** A text input field with a red border and a red error icon. Below it, the message "Password must be at least 8 characters." is displayed in red.
- Australian Resident?** A checkbox labeled "Australian Resident?". Below it, the message "Please confirm that you are an Australian resident." is displayed in red.
- Reason For Joining:** A text area with a red border and a red error icon. Below it, the message "Please enter at least 10 characters." is displayed in red.
- Gender:** A dropdown menu with a red border and a red error icon. Below it, the message "Please select a gender." is displayed in red.

At the bottom of the form, there are two buttons: "Submit" (blue) and "Clear" (grey). Below the buttons, there is a small icon of a person with a red error icon.

Figure 8. Five validation messages after submitting an empty form.

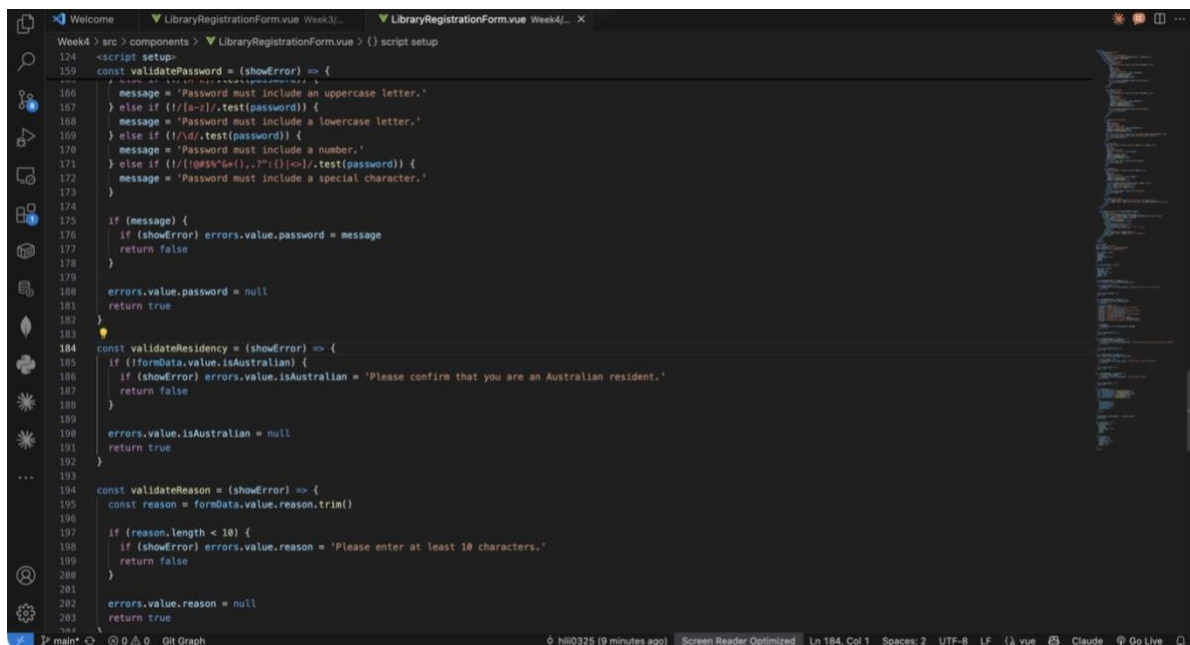
6. Week 4 - Validation Code

The validation functions return true or false. They also place the message in the errors object when an input is invalid.



```
Week4 > src > components > LibraryRegistrationForm.vue > {} script setup
124 <script setup>
125
126
127
128 const formData = ref({
129   username: '',
130   password: '',
131   isAustralian: false,
132   reason: '',
133   gender: ''
134 })
135
136
137 const submittedUsers = ref([])
138
139 const errors = ref({
140   username: null,
141   password: null,
142   isAustralian: null,
143   reason: null,
144   gender: null
145 })
146
147 const validateUsername = (showError) => {
148   const username = formData.value.username.trim()
149
150   if (username.length < 3) {
151     if (showError) errors.value.username = 'Username must be at least 3 characters.'
152     return false
153   }
154   errors.value.username = null
155   return true
156 }
157
158 const validatePassword = (showError) => {
159   const password = formData.value.password
160   let message = null
161
162   if (password.length < 8) {
163     message = 'Password must be at least 8 characters.'
164   } else if (!/[A-Z]/.test(password)) {
165     message = 'Password must include an uppercase letter.'
166   }
167   if (message) {
168     if (showError) errors.value.password = message
169     return false
170   }
171   errors.value.password = null
172   return true
173 }
174
175 const validateResidency = (showError) => {
176   if (!formData.value.isAustralian) {
177     if (showError) errors.value.isAustralian = 'Please confirm that you are an Australian resident.'
178     return false
179   }
180   errors.value.isAustralian = null
181   return true
182 }
183
184 const validateReason = (showError) => {
185   const reason = formData.value.reason.trim()
186
187   if (reason.length < 10) {
188     if (showError) errors.value.reason = 'Please enter at least 10 characters.'
189     return false
190   }
191   errors.value.reason = null
192   return true
193 }
```

Figure 9. Username and password validation code.

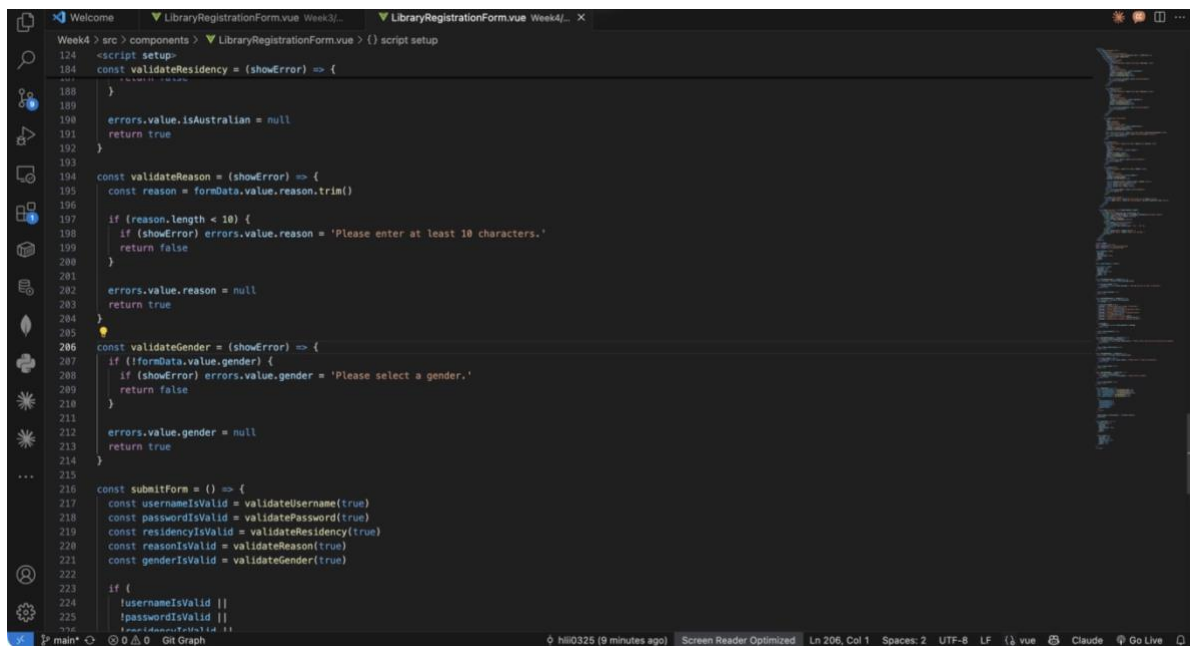


```
Week4 > src > components > LibraryRegistrationForm.vue > {} script setup
124 <script setup>
125
126
127
128 const formData = ref({
129   username: '',
130   password: '',
131   isAustralian: false,
132   reason: '',
133   gender: ''
134 })
135
136
137 const submittedUsers = ref([])
138
139 const errors = ref({
140   username: null,
141   password: null,
142   isAustralian: null,
143   reason: null,
144   gender: null
145 })
146
147 const validateUsername = (showError) => {
148   const username = formData.value.username.trim()
149
150   if (username.length < 3) {
151     if (showError) errors.value.username = 'Username must be at least 3 characters.'
152     return false
153   }
154   errors.value.username = null
155   return true
156 }
157
158 const validatePassword = (showError) => {
159   const password = formData.value.password
160   let message = null
161
162   if (password.length < 8) {
163     message = 'Password must be at least 8 characters.'
164   } else if (!/[A-Z]/.test(password)) {
165     message = 'Password must include an uppercase letter.'
166   } else if (!/[a-z]/.test(password)) {
167     message = 'Password must include a lowercase letter.'
168   } else if (!/[0-9]/.test(password)) {
169     message = 'Password must include a number.'
170   } else if (!/[!@#$%^&*()_-=+{}|~`'"/>
171     message = 'Password must include a special character.'
172   }
173   if (message) {
174     if (showError) errors.value.password = message
175     return false
176   }
177   errors.value.password = null
178   return true
179 }
180
181 const validateResidency = (showError) => {
182   if (!formData.value.isAustralian) {
183     if (showError) errors.value.isAustralian = 'Please confirm that you are an Australian resident.'
184     return false
185   }
186   errors.value.isAustralian = null
187   return true
188 }
189
190 const validateReason = (showError) => {
191   const reason = formData.value.reason.trim()
192
193   if (reason.length < 10) {
194     if (showError) errors.value.reason = 'Please enter at least 10 characters.'
195     return false
196   }
197   errors.value.reason = null
198   return true
199 }
```

Figure 10. Residency and reason validation code.

7. Final Validation Check (Week 4)

The submit function calls all five validation functions. The form only adds the user when every result is valid.

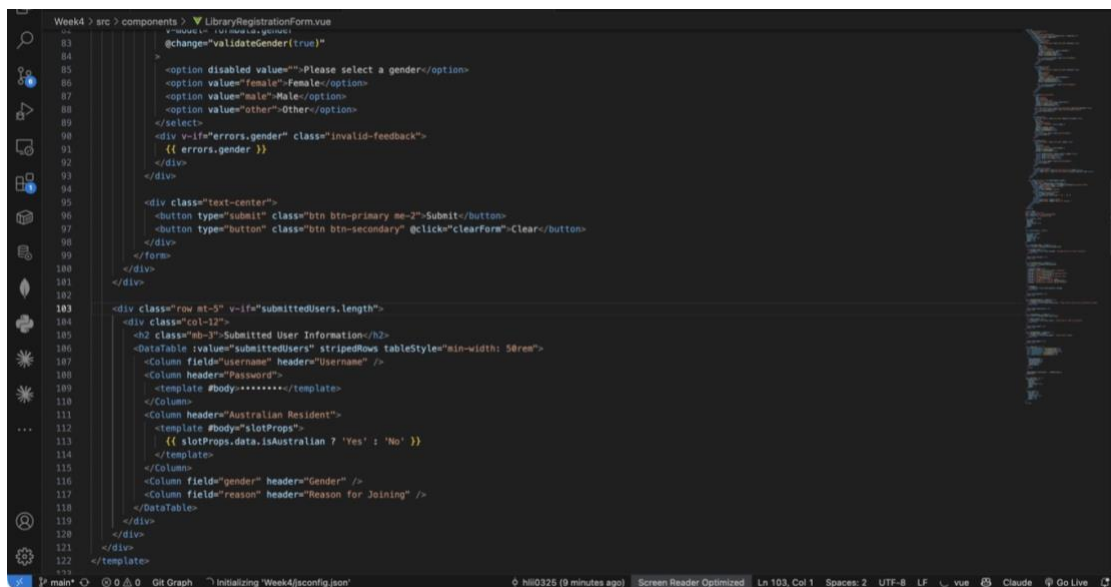


```
Week4 > src > components > LibraryRegistrationForm.vue > {} script setup
124 <script setup>
184 const validateResidency = (showError) => {
185   // ...
188 }
189
190 errors.value.isAustralian = null
191 return true
192 }
193
194 const validateReason = (showError) => {
195   const reason = formData.value.reason.trim()
196
197   if (reason.length < 10) {
198     if (showError) errors.value.reason = 'Please enter at least 10 characters.'
199     return false
200   }
201
202   errors.value.reason = null
203   return true
204 }
205
206 const validateGender = (showError) => {
207   if (formData.value.gender) {
208     if (showError) errors.value.gender = 'Please select a gender.'
209     return false
210   }
211
212   errors.value.gender = null
213   return true
214 }
215
216 const submitForm = () => {
217   const usernameIsValid = validateUsername(true)
218   const passwordIsValid = validatePassword(true)
219   const residencyIsValid = validateResidency(true)
220   const reasonIsValid = validateReason(true)
221   const genderIsValid = validateGender(true)
222
223   if (
224     !usernameIsValid ||
225     !passwordIsValid ||
226     !residencyIsValid ||
227     !reasonIsValid ||
228     !genderIsValid
229   ) {
230     // ...
231   }
232 }
233
```

Figure 11. Gender validation and the five checks in submitForm.

8. Week 4 - PrimeVue DataTable

I used PrimeVue DataTable and Column components to show submitted users. The password is masked in the table instead of displaying the real value.



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Week4 > src > components > LibraryRegistrationForm.vue
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